

Incentive mechanisms to enforce sustainable forest exploitation

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Using a two-player differential game approach, this paper deals with the issue of tropical deforestation. The assumption is that developing forestry countries (called the *South*) do not, or cannot, necessarily have the same utility for forest conservation than the developed countries (called the *North*). Given the global nature of the problem, we assume that it may be wise that the *North* offers financial support to the *South* to help in achieving a sustainable forest policy. We derive sustainable and short-run deforestation policies and provide a transfer mechanism which insures that the *South* will indeed implement the sustainable one.

Keywords: tropical deforestation, sustainable policy, incentive mechanism, differential games

1. Introduction

The global rate of tropical forest destruction is raising general concern. According to the Food and Agriculture Organization's (FAO) estimates, 53,000 mi² of tropical forests were destroyed each year during the 1980s. A report of the World Resources Institute (see Matthews [18]) confirms that the deforestation rate is not slowing, but, on the contrary, continues to be rapid.

The adverse consequences of tropical deforestation on biodiversity conservation and climate change are considerable. In fact, the rate of carbon dioxide released in the atmosphere due to tropical deforestation is approximately 1.6 billion metric tons per year. This constitutes a significant contribution compared to a rate of 6 billion metric tons of carbon emitted from fossil fuel burning, one of the main factors of global warming [9]. On the other hand, serious scientific estimates indicate that, on average, 137 species forms of life (plants and animals) are driven into extinction every day due to habitat destruction [27].

The main causes of tropical deforestation seem to be the conversion of forested land to agricultural use and, to a lower level, the forestry activities (e.g., [1–3, 16, 21, 22]).

It is easily understood that since a country gets revenue from agriculture and forestry activities, the temptation is high to follow a *laissez-faire* policy when it comes to deforestation. In fact, scholars are pointing out that even if the costs of forest preservation are small compared to the large non-economic benefits from doing so, at a domestic level these benefits remain much smaller than the global ones (e.g., [7, 8, 20]). Barret [5], Von Amsberg [26], Van Soest [24] and Van Soest and Lensink [25] argue that in

some instances the allocation of forest lands to alternative use may enhance the domestic country's welfare, while decreasing the welfare from the global perspective. This confers to deforestation its international externality dimension. One possibility to tackle this problem by rich countries which value the tropical forest is to compensate financially those forestry countries which accept to implement conservation policies.

Given the dynamic nature of this problem, the literature has adopted differential games and optimal control theory formalisms to design financial transfers from the developed countries (termed the *North*) to forestry developing countries (called the *South*). Barbier and Rauscher [4], using optimal control theory, consider lump-sum aid donations as an indirect instrument to conserve the forest by reducing the necessity to exploit it. However, lump-sum transfers have been criticized as being a passive instrument to combat deforestation. A more active way to use transfers would be to make the amount of payment conditional to the effort deployed by the recipient country to conserve the rainforest.

In this perspective, a transfer that consists in paying a fixed price per unit of land conserved was proposed by Stähler [23], who used an optimal control approach, and by Mohr [19], in a bargaining game framework. These authors raise the point that this kind of dependency between the transfers and the forest size may, however, have adverse effects in the long run. As the international community's willingness to pay is higher when the forested land becomes smaller, the forestry country can use its "market power" to raise the per-unit compensation through a strategic deforestation behavior. Mohr [19] explained that, in this case, the credibility problem about whether the donor community is indeed "hard nosed" about the fixed compensation can give an incentive to the recipient country to increase its deforestation rate.

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Consequently, a transfer payment that penalizes an excessive deforestation rate in addition to compensating for each unit of conserved forested land may be advantageous, especially if we consider that the ecosystem biodiversity is, to a large extent, irreversible. Such a transfer payment, which makes the recipient countries directly confronted to the results of their land-use decisions, was first used in a contract approach by Van Soest and Lensink [25], and in a differential game approach by Fredj et al. [14] and Martín-Herrán et al. [17].

In all these models, the *North's* main concern is forest preservation, whereas the forestry country's objective is the maximization of the total revenues it can extract from forestry and agricultural activities. In other words, the problem here is essentially that resource managers do not internalize the loss of biodiversity or the loss of an important carbon sink in their objective function. What this means is that the benefit of tropical forests in terms of cure from cancer and other illnesses and the mitigation of climate change, which represent the concerns of the *North*, ignored by the domestic forestry countries.

Consequently and as it has been proven in the above studies, it should be in the interest of both the *North* and the *South* to implement the agreed transfer program. Nevertheless, there is no guarantee that both players would stick to their respective engagements as dictated by the agreed policy. One of the players can have the advantage and hence the incentive to deviate from the initial agreement if it knows that the other player will stick to its engagement. The strategies implied by the agreement are then no longer an equilibrium.

This calls for the use of incentive mechanism strategies to counter this eventual problem. The externality problem in several environmental issues have led to the use of incentive mechanism design, e.g., Jørgensen and Zaccour [15] use incentive equilibrium strategies in the pollution control problem. Originally the idea of incentive equilibrium has been developed and applied to resource management problems by Ehtamo and Hämäläinen [10–12].

This paper is concerned with the design of incentive strategy mechanisms that can be used by a donor community so that the forestry countries find it optimal to choose in equilibrium a deforestation policy that is sustainable in the long run. We assume that the Northern community has a real concern about the forest preservation. Therefore, it can not put its future credibility in jeopardy by retracting at an intermediate date from what it announced at the initial instant of the game, especially considering that this environmental agreement is expected to be renewed at the end of the horizon or later on. We then exclude the possibility of deviation from this player. However, unilateral deviation remains possible for the forestry country as it can be more beneficial for it to receive the compensating transfer without making any effort of forest conservation. To avoid this problem, the *North* can use an incentive mechanism to force the forestry country to stick to the agreed strategy.

The objective of the Northern community is that, while maximizing its stream of revenues, the forestry country chooses a deforestation rate that is sustainable in the long run. The official definition of sustainable development stated in the World Commission on Environment and Development's report [28] is "development that meets the needs of the present without compromising the ability of future generations to meet their own needs."

To ensure the participation of the forestry country in the program, the agreement should ensure at least the same actualized revenues as what the forestry country can have without the developed community's intervention, for the period covered by the agreement. Furthermore, to guarantee the implementation of the agreement once the forestry country accepts to receive the transfer, the *North* has to make the transfer dependent directly on the forestry country's actions regarding the forest exploitation. The objective from doing so is that the forestry country finds it optimal to choose in the short-run equilibrium the deforestation rate that leads to sustainable forest stock in the long run. This happens when, for instance, the deforestation rate is equal to the forest's natural regeneration (see, for example, [6]). Our results show that this is possible and can be implemented using an incentive transfer mechanism where the amount of transfers is linear in the forest stock and linear-quadratic in the deforestation rate.

The main contributions of the paper are the following. First, we solve for both the short-run and long-run equilibria. This allows us to calculate the sustainable forest exploitation in the long run and to estimate the loss in the total welfare the forestry country can bear from a more sustainable exploitation.

Second, we define a transfer function which is composed of a lump-sum amount, which guarantees the participation of the recipient country while compensating it for the total loss due to a better forest conservation, and of a conditional part, which represents the incentive mechanism ensuring that the forestry country will respect its engagement regarding a sustainable forest exploitation. To our knowledge, this kind of incentive mechanism has never been used previously in the sustainable forest management literature.

In contrast with other previous studies, we do not need to explicitly model the developed community's utility function, which palliates one of the main criticisms addressed in this literature. Moreover, we consider different dynamics of the forest stock as we allow the possibility of regeneration of the resource.

The rest of the paper is organized as follows. Section 2 introduces the model. Section 3 characterizes the optimal control solutions for the forestry country without the intervention of the Northern community, under the finite and infinite planning horizons. Section 4 deals with the issue of distribution over time of the total payment flowing from *North* to *South* and with the design of an incentive mechanism that guarantees the application of the sustain-

able forest exploitation at each time t . Section 5 provides a numerical illustration of our results. Finally, section 6 concludes.

2. The model

The original model is proposed for the forestry country by Ehui et al. [13] as an allocation problem of forested land between forest activities and agricultural use. The objective is to maximize the present value of a utility index $U(\cdot)$, which measures society's satisfaction. The utility index is concave increasing in the aggregate benefit (π), measuring the societal benefit from agriculture and forestry. The latter depends mainly on the forest size, the deforestation rate and the soil's productivity.

A detailed and sophisticated specification of this model is proposed by Van Soest and Lensink [25], where the utility index is presented as the total net revenues from agricultural and forest use.

The revenues from forest activities are measured at each time t as the wood production $q(t)$ times the prevailing price of timber $P(t)$. The timber can be extracted using clearfelling or selective logging methods. Under the first method, the land is completely deforested and converted to agricultural use. Assuming that there are n commercially valuable stems per unit of land, the quantity of timber supplied at each period of time t is then equal to n times the deforestation rate $D(t)$. Under the selective logging method, only a fraction γ of timber is extracted and the quantity supplied at each period of time t is equal to γ times n multiplied by the size of the forest $F(t)$. The total timber production can be described as follows:¹

$$q(t) = nD(t) + n\gamma F(t). \quad (1)$$

For convenience, n is normalized to unity such that the timber price $P(t)$ represents the value of all commercially valuable timber per unit of land. The (inverse) demand function is assumed, following a long tradition in economics literature, linear and given by

$$P(t) = \bar{P} - \theta q(t), \quad (2)$$

where $\bar{P}$ is the maximal market price obtained when $D(t)$ tends toward zero and θ is a positive parameter.

The agricultural net revenue depends on the size of the land under cultivation $F_0 - F(t)$, where F_0 is the initial size of the forest stock; on the agricultural price which is fixed to $\bar{P}_A$; and on land productivity $Z(t)$. The latter depends positively on the deforestation rate since burning off the

forest cover releases nutrients that increase the soil productivity in the short run. It is decreasing in the cumulative deforestation as, in the long run, soil productivity falls because of nutrient depletion and decreases because the increasing distance of forest cover causes the deceleration of the soil formation:

$$Z(t) = \bar{Z} + \alpha D(t) - \beta[F_0 - F(t)]. \quad (3)$$

Combining the revenues from forest exploitation and agriculture activities the total revenue function of the *South* is thus

$$R(t) = P(t)q(t) + \bar{P}_A Z(t)[F_0 - F(t)]. \quad (4)$$

This function shows clearly that there is a trade-off between deforestation and conservation of tropical forests.

We suppose that the rainforest evolves according to the following differential equation typical in a renewable resource context,²

$$\dot{F}(t) = -D(t) + rF(t); \quad F(0) = F_0, \quad (5)$$

where r denotes the natural regeneration rate of the rainforest.

The forestry country choosing the deforestation rate aims at maximizing its stream of revenues discounted at rate ρ :

$$\int_0^T e^{-\rho t} R(t) dt + e^{-\rho T} \Phi(F(T)),$$

subject to (1) – (5),

where function Φ is the salvage value function. The time horizon $[0, T]$ may be bounded or unbounded depending on whether T is finite or infinite. In the infinite horizon case, the salvage value function is identically null.

This dynamic optimization problem belongs to the well known linear–quadratic class. Indeed, the motion equation of the state variable, the forest stock, is linear and the objective function is quadratic in both the state and control variables. The total revenue function of the *South* can be expressed as:

$$R(t) = -a_1 F(t)^2 + a_2 F(t) - a_3 D(t)^2 + a_4 D(t) - a_5 D(t)F(t) + a_6,$$

where

$$\begin{aligned} a_1 &= \beta \bar{P}_A + \theta \gamma^2, \quad a_2 = \gamma \bar{P} + \bar{P}_A (2\beta F_0 - \bar{Z}), \quad a_3 = \theta, \\ a_4 &= \bar{P} + \alpha \bar{P}_A F_0, \quad a_5 = 2\theta \gamma + \alpha \bar{P}_A, \quad a_6 = \bar{P}_A F_0 (\bar{Z} - \beta F_0). \end{aligned} \quad (6)$$

¹ In the original specification of the model by Van Soest and Lensink [25], the timber supply under the selective method is equal to $n\gamma(F(t) - D(t))$. As we are working with a dynamic model we find it more convenient to consider that the size of the forest at any time is equal to the size of land not converted to agriculture use.

² In the model specified by Van Soest and Lensink [25], the forest dynamics is typical of a non renewable resource with $\dot{F}(t) = -D(t)$.

This functional form shows how the forest stock and the deforestation rate affect the instantaneous revenues. $a_1 > 0$ (respectively $a_5 > 0$) reflect the decrease in the forest marginal revenues when the forest stock (respectively the deforestation) rate increases. $a_3 > 0$ (respectively $a_5 > 0$) reflect the decrease in the marginal revenues of deforestation when the deforestation rate (respectively the forest stock) increases.

3. Optimal forest exploitation

In this section, we derive optimal forest exploitation by the *South* assuming in turn finite and infinite horizons. We conjecture that the short-run (finite-horizon) optimization leads to an over extraction as the forestry country does not take into account the long-run impacts of its actions. For this reason, we consider as a counterpart the optimization problem with an infinite horizon, which leads in the long-run equilibrium to a sustainable exploitation with $\dot{F}(t) = 0$, implying that the deforestation rate at equilibrium in the long-run will be equal to the natural regeneration of the forest.

Our objective is to induce the forestry country to choose a sustainable deforestation rate while optimizing in the short run (or finite horizon). In a first step, we derive the solutions of the optimization problems with finite and infinite time horizons. Afterwards, in the next section, we will design the necessary incentive strategies that can be used by a second party, which could be the Northern community as addressed in the previous literature, that will enforce the sustainable forest exploitation in the forestry country.

3.1. Infinite Horizon

The optimization problem in the infinite horizon case reads as follows

$$\max_{\{D(t)\}} \int_0^\infty e^{-\rho t} R(t) dt,$$

subject to (1) – (5).

The following proposition provides the optimal solution to the *South* problem when an infinite planning horizon is considered. The superscript ‘s’ stands for *sustainable* solution.

PROPOSITION 1. Assuming an interior solution, the optimal control, state and costate variables in the infinite horizon scenario which leads to a forest stock path converging in the long-run to the steady state, F^* , are given by:

$$F^s(t) = F^* + (F_0 - F^*)e^{(\rho - \Delta)t/2},$$

$$\lambda^s(t) = \lambda^* + (F_0 - F^*)(a_3(\rho - 2r - \Delta) - a_5)e^{(\rho - \Delta)t/2}, \quad (7)$$

$$D^s(t) = rF^* - \frac{1}{2}(\rho - \Delta - 2r)(F_0 - F^*)e^{(\rho - \Delta)t/2}, \quad (8)$$

where F^* and λ^* represent, respectively, the long-run equilibrium or steady state of the forest size and its shadow value, given by the following expressions:

$$F^* = \frac{a_2 + a_4(r - \rho)}{2a_1 - 2a_3r(\rho - r) - a_5(\rho - 2r)}, \quad (9)$$

$$\lambda^* = \frac{-a_2(a_5 + 2ra_3) + a_4(ra_5 + 2a_1)}{2a_1 - 2a_3r(\rho - r) - a_5(\rho - 2r)}, \quad (10)$$

$$\text{and } \Delta = \sqrt{\frac{4a_1 + (\rho - 2r)[(\rho - 2r)a_3 - 2a_5]}{a_3}}.$$

The maximized net revenue per period of time is given by

$$R^s(t) = A_1e^{(\rho - \Delta)t} + A_2e^{(\rho - \Delta)t/2} + A_3,$$

where constants A_i for $i = 1, 2, 3$ are given in the [Appendix](#).

Proof. The current value Hamiltonian associated with the problem described above is³

$$\begin{aligned} H(D, F, \lambda) &= R(D, F) + \lambda \dot{F} \\ &= -a_1F^2 + a_2F - a_3D^2 + a_4D - a_5DF \\ &\quad + a_6 + \lambda(-D + rF), \end{aligned}$$

where λ is the costate variable associated with the forest stock F .

An interior solution must satisfy the first-order optimal conditions given by

$$H_D = -2a_3D - a_5F - \lambda + a_4 = 0, \quad (11)$$

$$\dot{F} = H_\lambda = -D + rF; \quad F(0) = F_0, \quad (12)$$

$$\dot{\lambda} = \rho\lambda - H_F = 2a_1F + a_5D + (\rho - r)\lambda - a_2, \quad (13)$$

together with the transversality condition $\lim_{t \rightarrow \infty} e^{-\rho t} \lambda(t) F(t) = 0$.

Since $H_{DD} = -2a_3 < 0$, the above conditions are also sufficient for this maximization problem.

Replacing parameters a_3 , a_4 and a_5 by their original values in terms of the model's parameters as defined in (6), and taking into account equation (1) after few manipulations we can rewrite condition (11) as:

$$\lambda = \bar{P} - 2\theta q + \alpha \bar{P}_A(F_0 - F).$$

³ For the rest of the paper, time argument is omitted when no confusion can arise.

The second condition (12) simply describes the dynamics of the forest, decreasing at the deforestation rate D , reduced by the natural regeneration of the forest.

From equation (11), we can also extract the expression of D as a function of the state variable F and its shadow value λ :

$$D = -\frac{1}{2a_3}(\lambda - a_4 + a_5F). \quad (14)$$

Replacing the latter in (12) and (13), we get the following system of differential equations:

$$\begin{aligned} \dot{F} &= \left(\frac{a_5}{2a_3} + r\right)F + \frac{1}{2a_3}\lambda - \frac{a_4}{2a_3}; \quad F(0) = F_0, \\ \dot{\lambda} &= \left(2a_1 - \frac{1}{2a_3}a_5^2\right)F + \left(\rho - r - \frac{a_5}{2a_3}\right)\lambda + \frac{a_4a_5}{2a_3} - a_2. \end{aligned}$$

Solving this dynamic system taking into account the transversality condition $\lim_{t \rightarrow \infty} e^{-\rho t} \lambda(t) F(t) = 0$, we have:

$$\begin{aligned} F^s(t) &= F^* + (F_0 - F^*)e^{(\rho - \Delta)\frac{t}{2}}, \\ \lambda^s(t) &= \lambda^* + (F_0 - F^*)(a_3(\rho - 2r - \Delta) - a_5)e^{(\rho - \Delta)\frac{t}{2}}, \end{aligned}$$

where F^* and λ^* respectively represent the long run equilibrium or steady state of the forest size and its shadow value which expressions are given in the statement of the proposition.

Substituting F and λ by their optimal values F^s and λ^s in (14), we then get the optimal path of the deforestation rate:

$$\begin{aligned} D^s(t) &= -\frac{1}{2a_3}(\lambda^* + a_5F^* - a_4) \\ &\quad + \frac{4a_1 - a_5(\Delta - 2r + \rho)}{2(a_3(\Delta - 2r + \rho) - a_5)}(F_0 - F^*)e^{(\rho - \Delta)\frac{t}{2}}, \end{aligned}$$

which after some manipulations can be rewritten as in (8).

In order to have the optimal forest stock path converging to the steady-state F^* , we need to impose that $\rho - \Delta < 0$. This condition is satisfied if and only if the following inequality applies:

$$2a_1 - 2a_3r(\rho - r) - a_5(\rho - 2r) > 0. \quad (15)$$

This condition guarantees that the term inside the square root in Δ is always positive. Moreover, under this condition the steady-state values of the forest size and its shadow value are positive if the following conditions are satisfied:

$$\begin{aligned} a_2 + a_4(r - \rho) &> 0, \\ a_4(2a_1 + ra_5) - a_2(a_5 + 2ra_3) &> 0. \end{aligned} \quad (16)$$

Replacing the deforestation and the forest stock by their optimal paths, D^s and F^s , in the objective function, the net revenues from forest exploitation and agriculture use can be written as

$$R^s(t) = A_1e^{(\rho - \Delta)t} + A_2e^{(\rho - \Delta)\frac{t}{2}} + A_3,$$

where constants A_i for $i = 1, 2, 3$ are given in the Appendix. $\square$

Of interest is the interpretation of the costate variable

$$\lambda = \bar{P} - 2\theta q + \alpha \bar{P}_A(F_0 - F).$$

The above expression means that the optimal deforestation rate at each time t should be such that the marginal cost of deforestation (measured by the current shadow value of the forest) equals the net marginal benefit (the sum of the marginal income from selling the timber $\bar{P} - 2\theta q$ and the additional agriculture revenue $\alpha \bar{P}_A(F_0 - F)$).

To interpret the variation over time of λ , we replace in (13) the parameters a_i ($i = 1, 2, 5$) by their original values given in (6), to obtain, after some arrangements,

$$\dot{\lambda} = (\rho - r)\lambda - \gamma[\bar{P} - 2\theta q] + \bar{P}_A[\bar{Z} + \alpha D - 2\beta(F_0 - F)].$$

This condition indicates that the shadow value of the forest increases or decreases at rate $\rho - r$, depending on the sign of $\rho - r$, reduced by the net benefits of conserving an additional unit of land. The latter is equal to the direct marginal revenues of logging it minus the opportunity cost, which is equal to the indirect marginal agriculture revenue that could be earned from conserving this unit of land which indirectly enhances agricultural yield through the productivity effect.

From now on we assume that $\rho - r < 0$, which guarantees the fulfillment of (15) and the first inequality in (16). The values of the model's parameters are considered such that the second inequality in (16) is satisfied.

The asymptotically stable trajectory for the forest stock increases or decreases towards its steady state depending on the initial size of the forest. Under condition $\rho - r < 0$, it is easy to see that the convergence of the forest shadow value to its steady state presents the opposite behavior. That is, if the forest stock increases its shadow value decreases and vice versa.

The steady state of the forest stock, equation (9), can also be written as

$$\begin{aligned} F^* &= \frac{\gamma \bar{P} + (\bar{P} + \alpha F_0 P_A)(r - \rho) + \bar{P}_A(2\beta F_0 - \bar{Z})}{2\beta \bar{P}_A + 2\theta \gamma^2 - 2r\theta(\rho - r) - (\rho - 2r)(2\theta \gamma + \alpha \bar{P}_A)} \\ &= F_0 - \frac{\bar{P}_A(\bar{Z} + \alpha r F_0) - (r - \rho + \gamma)(\bar{P} - 2(r + \gamma)\theta F_0)}{2\beta \bar{P}_A + 2\theta \gamma^2 - 2r\theta(\rho - r) - (\rho - 2r)(2\theta \gamma + \alpha \bar{P}_A)}. \end{aligned}$$

If $\rho - r < 0$, the denominator of the second term in the right-hand side is positive. The forest size will be decreasing ($F^* < F_0$) if and only if the numerator is also positive. The comparative statistics shows that higher forestry revenues induced by a relative increase in the timber price leads to a higher steady state of the forest stock. This tallies with the corresponding result in Ehui et al. [13]. An increase in the land productivity has a negative effect on the steady-state level of the forest size. The effect of a relative increase in agriculture price is,

however, undetermined and so is the effect of the discount rate.

3.2. Finite horizon

In a finite-horizon optimization problem, it is customary to add a salvage value to acknowledge that, although the planning horizon is short, something will still happen after the terminal date. Put differently, the salvage value represents the payoff-to-go starting at T . We shall assume for simplicity that this salvage value is linear in the stock of rainforest, i.e., $\Phi(F(T)) = \phi F(T)$. Clearly, this is an approximation since, in the language of dynamic programming, the value function precisely measuring the payoff-to-go of the problem is not linear. A drastic case occurs when ϕ is taken equal to zero, which would correspond to an after-me-the-deluge policy and a complete depletion of the rainforest.

The optimization problem to be solved is given by

$$\max_{\{D(t)\}} \int_0^T e^{-\rho t} R(t) dt + e^{-\rho T} \phi F(T),$$

subject to (1) – (5).

The following proposition provides the optimal solution to this problem, where the superscript ‘ T ’ stands for bounded finite horizon scenario, with terminal time T .

PROPOSITION 2. Assuming an interior solution, the short-run optimal control, state and costate variables are given by

$$F^T(t) = F^* + \frac{a_3(\Delta + \rho - 2r) - a_5}{a_5^2 - 4a_1a_3} C_1 e^{(\rho-\Delta)\frac{t}{2}} + \frac{(\rho - 2r - \Delta)a_3 - a_5}{a_5^2 - 4a_1a_3} C_2 e^{(\rho+\Delta)\frac{t}{2}}, \quad (17)$$

$$\lambda^T(t) = \lambda^* + C_1 e^{(\rho-\Delta)\frac{t}{2}} + C_2 e^{(\rho+\Delta)\frac{t}{2}}, \quad (18)$$

$$D^T(t) = \frac{1}{2a_3} (a_4 - \lambda^* - a_5 F^*) + \frac{[(\Delta + \rho - 2r)a_5 - 4a_1]C_1}{2(a_5^2 - 4a_1a_3)} e^{(\rho-\Delta)\frac{t}{2}} + \frac{[4a_1 - (\rho - 2r - \Delta)a_5]C_2}{2(a_5^2 - 4a_1a_3)} e^{(\rho+\Delta)\frac{t}{2}}. \quad (19)$$

The maximized net revenue per period t is given by

$$R^T(t) = B_1 e^{t\rho} + B_2 e^{t(\rho+\Delta)} + B_3 e^{(\rho+\Delta)\frac{t}{2}} + B_4 e^{(\rho-\Delta)t} + B_5 e^{(\rho-\Delta)\frac{t}{2}} + B_6.$$

Constants C_1 , C_2 and B_i , $i = 1, \dots, 6$ are given in the [Appendix](#).

Proof. The first-order optimal conditions are given by (11)–(13) together with the transversality condition $\lambda(T) = \phi$.

Following the same reasoning than in the infinite horizon scenario, the optimal solutions described in the proposition above can be derived.

Inserting the optimal time paths for the control and state variables, D^T and F^T , in the objective function gives the optimal revenue R^T . $\square$

The results in the above proposition are intuitive. Indeed, it is readily seen from the optimal conditions that the deforestation policy satisfies the familiar rule of marginal revenue from deforestation must equal its marginal cost, as in the infinite horizon maximization problem. The interpretation of the results stays the same, except that in the finite problem the forest stock path does not converge to the long-run equilibrium where the forest exploitation is sustainable.

4. Incentive strategies

An assumption in our framework is that the *North* or donor community is willing to compensate the forestry country for the loss in welfare it would incur when following the sustainable strategy for as long as it has the insurance that the *South* will indeed implement the desired conservation policy. To design an incentive mechanism that allows us to achieve our objective, we use the following *algorithm*:

1. We first determine the total amount of the transfer by computing the difference between the *South*'s payoff under the short-run and long-run (sustainable) scenarios. Given that the support program is assumed to be in place over the interval $[0, T]$, an assumption must be made on how the *South* would value the forest stock at T . We assume that it uses the same salvage function as in the short-run horizon and thus the total amount to be transferred from the *North* to the *South* is given by

$$J^T - J^s = \int_0^T e^{-\rho t} (R^T(t) - R^s(t)) dt + e^{-\rho T} \phi (F^T(T) - F^s(T)).$$

Note that the above quantity is interpreted as a minimal requirement to induce the *South* to choose the sustainable conservation policy.

2. We second assume that the *North* distribute the total amount over time as an incentive mechanism, that is by letting, as in Van Soest and Lensink [25] and Fredj et al. [14], the instantaneous transfer be dependent on both the deforestation rate and the forest stock, i.e.,

$$S^I(t) = S(D, F), \quad (20)$$

where the superscript ‘ I ’ stands for *incentive* mechanism. We assume that $S(D, F)$ is increasing in F and decreasing in D .

Remark 1. There are many options to define the instantaneous transfer. Indeed, any time-function $S(t)$ satisfying the requirement that the sum of discounted stream of transfers equals the total amount, i.e.,

$$\int_0^T e^{-\rho t} S(t) dt = J^T - J^s, \quad (21)$$

is a priori a candidate. To illustrate, $S(t)$ can be taken as the difference in instantaneous revenues, reduced by the present value of the average difference in the forest salvage value at the end of the horizon, i.e.,

$$S(t) = (R^T(t) - R^s(t)) + \frac{e^{-\rho(T-t)} \phi[F^T(T) - F^s(T)]}{T},$$

Another option is to have a constant transfer over time:

$$\bar{S} = \frac{\rho}{1 - e^{-\rho T}} (J^T - J^s), \forall t \in [0, T].$$

Although the above two transfers satisfy (21), they do not guarantee that the *South* will indeed implement the desired sustainable strategy. That is why a transfer program must depend on the deforestation effort and eventually on the size of the forest is needed.

3. We last deal with the determination of a functional form and the parameters’ values for the incentive mechanism. Our approach is to redefine the optimization problem of the *South*, which includes now the transfers and obtain its deforestation strategy as a *best response* to $S(D, F)$, which can be specified afterwards. The new optimization problem of the *South* which includes now the transfer is

$$\max_{\{D(t)\}} \int_0^T e^{-\rho t} (R(t) + S(D, F)) dt + e^{-\rho T} \phi F(T),$$

subject to (1) – (5)

We now implement our algorithm. The next proposition provides the difference in welfare between the two scenarios.

PROPOSITION 3. The difference between the forestry country’s welfare for the finite and infinite planning time horizons is given by:

$$\begin{aligned} J^T - J^s = & TB_1 + \frac{B_2}{\Delta} (e^{\Delta T} - 1) + \frac{2B_3}{\Delta - \rho} (e^{-(\rho - \Delta)\frac{T}{2}} - 1) \\ & - \frac{M_1}{\Delta} (e^{-\Delta T} - 1) - \frac{2M_2}{\Delta + \rho} (e^{-(\rho + \Delta)\frac{T}{2}} - 1) \\ & + \phi \left[\frac{(\rho - 2r - \Delta)a_3 - a_5}{a_3^2 - 4a_1a_3} C_2 e^{-(\rho - \Delta)\frac{T}{2}} \right. \\ & \left. + \left(\frac{[a_3(\Delta + \rho - 2r) - a_5]C_1}{a_3^2 - 4a_1a_3} - (F_0 - F^*) \right) e^{-(\rho + \Delta)\frac{T}{2}} \right], \end{aligned} \quad (22)$$

where constants M_1 and M_2 are given in the [Appendix](#).

Proof. The forestry country’s welfare for the period $[0, T]$, under the optimal policies of forest exploitation associated to the finite and infinite horizon scenarios are respectively

$$\begin{aligned} J^T &= \int_0^T e^{-\rho t} R^T(t) dt + e^{-\rho T} \phi F^T(T), \\ J^s &= \int_0^T e^{-\rho t} R^s(t) dt + e^{-\rho T} \phi F^s(T). \end{aligned}$$

Therefore, the difference is given by:

$$\begin{aligned} J^T - J^s &= \int_0^T e^{-\rho t} (R^T(t) - R^s(t)) dt \\ &\quad + e^{-\rho T} \phi (F^T(T) - F^s(T)). \end{aligned}$$

After some easy computations, the following expression has been derived:

$$\begin{aligned} R^T(t) - R^s(t) &= B_1 e^{t\rho} + B_2 e^{t(\rho + \Delta)} + B_3 e^{(\rho + \Delta)\frac{t}{2}} \\ &\quad + M_1 e^{(\rho - \Delta)t} + M_2 e^{(\rho - \Delta)\frac{t}{2}}, \end{aligned} \quad (23)$$

where constants B_i , $i = 1, \dots, 3$ and M_j , $j = 1, 2$ are given in the [Appendix](#).

The difference between the size of the forest stock at time T under the finite and infinite time horizon scenarios is:

$$\begin{aligned} F^T(T) - F^s(T) &= \left(\frac{[a_3(\Delta + \rho - 2r) - a_5]C_1}{a_3^2 - 4a_1a_3} \right. \\ &\quad \left. - (F_0 - F^*) \right) e^{(\rho - \Delta)\frac{T}{2}} \\ &\quad + \frac{[a_3(\rho - 2r - \Delta) - a_5]C_2}{a_3^2 - 4a_1a_3} e^{(\rho + \Delta)\frac{T}{2}}. \end{aligned}$$

Thus, the integration of the difference in (23) over the time interval $[0, T]$ and the addition of the term $e^{-\rho T} \phi (F^T(T) - F^s(T))$ leads to the expression in (22). $\square$

The next proposition characterizes the solution of the new optimization problem of the *South*, i.e., the one including the transfers as revenues.

PROPOSITION 4. Assuming interior solutions, the optimal control, state and costate variables of the optimization problem when the forestry country receives a transfer function as in (20) satisfy:

$$D^I = \frac{1}{2a_3} \left(\frac{\partial S}{\partial D} (D^I, F^I) - a_5 F^I - \lambda^I + a_4 \right), \quad (24)$$

$$\dot{F}^I = -D^I + rF^I; \quad F^I(0) = F_0, \quad (25)$$

$$\dot{\lambda}^I = a_5 D^I + 2a_1 F^I + (\rho - r)\lambda^I - \frac{\partial S}{\partial F}(D^I, F^I) - a_2;$$

$$\lambda^I(T) = \phi. \quad (26)$$

Proof. The present value of the Hamiltonian associated with the new optimization problem is given by:

$$H^I = -a_1 F^2 + a_2 F - a_3 D^2 + a_4 D - a_5 D F + S(D, F) + \lambda^I(-D + rF) + \bar{S} + a_6,$$

where λ^I denotes the new costate variable associated with the forest stock.

Assuming interior solution the necessary conditions of the Maximum Principle of Pontryagin that guarantee the maximization of the new discounted stream of revenues are given by (24)–(26). $\square$

The incentive mechanism that guarantees the implementation of the sustainable exploitation path is such that F^s , the forest stock path for the infinite horizon optimization problem, given in (7), is the solution of the system (24)–(26).

The functional form of the transfer does not really matter for as long as it fulfills its job of ensuring implementability of the sustainable strategy. Therefore, easiness of computation becomes an important argument in choosing this form. If we take $S(F, D)$ linear, i.e.,

$$S(D, F) = v(t)F(t) - w(t)D(t), \quad (27)$$

where $v(t)$ and $w(t)$ are nonnegative functions, then the dynamic system (25)–(26) can be expressed, once the expression of D in (24) has been replaced, as a linear system in F and λ . The latter can be expressed in matrix form as follows:

$$\begin{bmatrix} \dot{F} \\ \dot{\lambda} \end{bmatrix} = A^I(t) \begin{bmatrix} F \\ \lambda \end{bmatrix} + B^I(t), \quad (28)$$

with boundary conditions

$$F(0) = F_0 \quad \text{and} \quad \lambda(T) = \phi, \quad (29)$$

where both the matrix $A^I(t)$ and the vector $B^I(t)$ depend on the specification and parameters of $S(D, F)$. To have as optimal path for the state variable the sustainable solution, we impose that

$$F^I(t) = F^s(t) = F^* + (F_0 - F^*)e^{(\rho - \Delta)\frac{t}{2}}, \quad \forall t \in [0, T],$$

where $F^I(t)$ denotes the forest stock path obtained as a solution of system (28).

The following proposition shows the existence of positive coefficients of $F(t)$ and $D(t)$ in the transfer function.

PROPOSITION 5. There exist positive functions $v(t)$ and $w(t)$ in $[0, T]$ which induce the forestry country to choose in

the short-run a deforestation rate leading to a sustainable forest path.

Proof. Our aim is to prove the existence of functions $v(t)$ and $w(t)$, such that

$$v(t) \geq 0, \quad w(t) \geq 0, \quad \forall t \in [0, T],$$

and implying that the sustainable deforestation path $D^s(t)$ solves the following optimal control problem:

$$\max_{D(t)} \int_0^T e^{-\rho t} [R(D(t), F(t)) + v(t)F(t) - w(t)D(t)] dt + e^{-\rho T} \phi F(T),$$

$$\text{s.t.: } \dot{F}(t) = -D(t) + rF(t), \quad F(0) = F_0,$$

$$\int_0^T e^{-\rho t} [v(t)F(t) - w(t)D(t)] dt = J^T - J^s.$$

The isoperimetric constraint can be written equivalently as:

$$\begin{aligned} \dot{z}(t) &= e^{-\rho t} [v(t)F(t) - w(t)D(t)]; \quad z(0) = 0; \\ z(T) &= J^T - J^s, \end{aligned} \quad (30)$$

where $\dot{z}(t)$ represents the transfer rate at t and $z(t)$ is the cumulative transfer from the start of the program till time t .

The current value Hamiltonian associated with the problem described above is

$$H^I = -a_1 F^2 + a_2 F - a_3 D^2 + a_4 D - a_5 D F + a_6 + vF - wD + \lambda^I(-D + rF) + \mu(vF - wD),$$

where λ^I, μ denote the new costate variables associated with the forest stock and the transfer constraint (30), respectively.

Assuming interior solution, the necessary conditions of the Maximum Principle that guarantee the maximization of the new stream of revenues are given by:

$$\begin{aligned} D^I &= \frac{1}{2a_3} (a_4 - a_5 F^I - \lambda^I - w(1 + \mu)), \\ \dot{F}^I &= -D^I + rF^I; \quad F^I(0) = F_0, \\ \dot{\lambda}^I &= a_5 D^I + 2a_1 F^I + (\rho - r)\lambda^I - v(1 + \mu) - a_2; \\ \lambda^I(T) &= \phi, \\ \dot{z} &= e^{-\rho t} (vF^I - wD^I); \quad z(0) = 0, \quad z(T) = J^T - J^s, \\ \dot{\mu} &= 0. \end{aligned} \quad (31)$$

From last differential equation we have that the costate variable μ is constant along the time horizon. Let us denote this constant by $\bar{\mu}$.

The incentive mechanism guaranteeing the implementation of the sustainable exploitation path is such that D^s and F^s , the deforestation and forest stock paths for the infinite horizon optimization problem, given in (8) and (7), satisfy the optimality conditions above.

From $D^I(t) = D^S(t) \forall t \in [0, T]$ we derive:

$$w(t)(1 + \bar{\mu}) = \lambda^S(t) - \lambda^I(t) \quad \forall t \in [0, T]. \quad (32)$$

Subtracting the differential equations that describe the time evolution of the costate variables we have:

$$\dot{\lambda}^S(t) - \dot{\lambda}^I(t) = (\rho - r)(\lambda^S(t) - \lambda^I(t)) + v(t)(1 + \bar{\mu}).$$

Differentiating with respect to time in (32) and replacing the expression of $\dot{\lambda}^S(t) - \dot{\lambda}^I(t)$, we have:

$$\dot{w}(t)(1 + \bar{\mu}) = (\rho - r)w(t)(1 + \bar{\mu}) + v(t)(1 + \bar{\mu}).$$

Let assume that $1 + \bar{\mu} \neq 0$, then last differential equation reads:⁴

$$\dot{w}(t) = (\rho - r)w(t) + v(t). \quad (33)$$

Equation (33) gives $v(t) = \dot{w}(t) - (\rho - r)w(t)$. Substituting this expression in the differential equation (31), where F^I and D^I have been replaced by F^S and D^S , respectively, we have:

$$\begin{aligned} \dot{z}(t) &= e^{-\rho t}[(\dot{w}(t) - (\rho - r)w(t))F^S(t) - w(t)D^S(t)] \\ &= e^{-\rho t}\dot{w}(t)F^S(t) - e^{-\rho t}\rho F^S(t)w(t) + e^{-\rho t}rF^S(t)w(t) \\ &\quad - e^{-\rho t}w(t)D^S(t) \\ &= e^{-\rho t}\dot{w}(t)F^S(t) - e^{-\rho t}\rho F^S(t)w(t) + e^{-\rho t}w(t)\dot{F}^S(t) \\ &= e^{-\rho t}\dot{w}(t)F^S(t) + w(t)\frac{d}{dt}(e^{-\rho t}F^S(t)) \\ &= \frac{d}{dt}[w(t)e^{-\rho t}F^S(t)]. \end{aligned}$$

Therefore,

$$z(t) = w(t)e^{-\rho t}F^S(t) + k_1,$$

where k_1 is a constant. The initial condition $z(0) = 0$ fixes the value of constant $k_1 = -w(0)F_0$, and then,

$$z(t) = w(t)e^{-\rho t}F^S(t) - w(0)F_0.$$

The final condition $z(T) = J^T - J^S$ leads to the following relationship:

$$w(T)e^{-\rho T}F^S(T) - w(0)F_0 = J^T - J^S. \quad (34)$$

⁴ Note that if $1 + \bar{\mu} = 0$, equation (32) implies:

$$\lambda^S(t) = \lambda^I(t) \quad \forall t \in [0, T].$$

Furthermore, the coefficient functions $v(t)$, $w(t)$ of the transfer function do not appear in the optimality conditions. Therefore, the donor community cannot select these functions in order to induce the forestry country to follow a sustainable exploitation of the forest. The donor community attains its objective only if $\lambda^S(T) = \phi$.

From (32), we obtain:

$$w(T) = \frac{\lambda^S(T) - \lambda^I(T)}{1 + \bar{\mu}} = \frac{\lambda^S(T) - \phi}{1 + \bar{\mu}}. \quad (35)$$

Replacing this expression in (34), we derive:

$$w(0) = \frac{1}{F_0} \left(J^S - J^T + \frac{\lambda^S(T) - \phi}{1 + \bar{\mu}} e^{-\rho T} F^S(T) \right). \quad (36)$$

Therefore, we have to solve the differential equation (33) taking into account the initial and final conditions given in (36) and (35). Let us note that both conditions depend on the value of the parameter $\bar{\mu}$. To show this dependence, let us denote the initial and final conditions by $w_0(\bar{\mu})$, $w_T(\bar{\mu})$.

The solution of the differential equation (33) satisfying the initial condition is given by:

$$w(t) = w_0(\bar{\mu})e^{(\rho-r)t} + \int_0^t e^{(\rho-r)(t-s)}v(s)ds. \quad (37)$$

The final condition implies the following relationship:

$$w_T(\bar{\mu}) = w_0(\bar{\mu})e^{(\rho-r)T} + \int_0^T e^{(\rho-r)(T-s)}v(s)ds.$$

From this expression, we get the value of parameter $\bar{\mu}$, which satisfies:

$$1 + \bar{\mu} = \frac{(\lambda^S(T) - \phi)(F_0 - F^S(T)e^{-rT})}{(J^S - J^T)e^{(\rho-r)T} + F_0 \int_0^T e^{(\rho-r)(T-s)}v(s)ds}, \quad (38)$$

where function $v(t)$ has to be selected such that the denominator of the expression above is not null.

From (37), it is straightforward to conclude that for any selection of $v(t)$ positive, if the initial condition $w_0(\bar{\mu})$ is non-negative, then $w(t) \geq 0 \forall t \in [0, T]$.

From (36), one has that $w_0(\bar{\mu})$ is non-negative if and only if

$$J^S - J^T + \frac{\lambda^S(T) - \phi}{1 + \bar{\mu}} e^{-\rho T} F^S(T) \geq 0. \quad (39)$$

Inequality (39) once the expression of $1 + \bar{\mu}$ given in (38) has been replaced establishes one constraint for the choice of function $v(t)$. This function has to be selected such that the following expressions have the same sign:

$$F_0 - F^S(T)e^{-rT},$$

$$J^S - J^T + e^{-\rho T} F^S(T) \int_0^T e^{(\rho-r)(T-s)}v(s)ds.$$

The differential equation (26) can be rewritten as:

$$\dot{\lambda}^I(t) = (\rho - r)\lambda^I(t) + H(t, \bar{\mu}), \quad \lambda^I(T) = \phi,$$

where

$$H(t, \bar{\mu}) = 2a_1 F^s(t) - a_2 + a_5 D^s(t) - v(t)(1 + \bar{\mu}).$$

The solution of this initial value problem is given by:

$$\lambda^I(t) = \phi e^{-(\rho-r)(T-t)} - \int_t^T e^{-(\rho-r)(s-t)} H(s, \bar{\mu}) ds.$$

□

The proposition shows that one can design a transfer function such that the forestry country would follow in the short term the sustainable deforestation policy. Note that we do not constrain the transfer $S(D, F)$ to be nonnegative at each instant of time. In the next example, $S(D, F)$ is positive for all t .

5. Numerical illustration

To have a better insight into the results (deforestation strategies, transfers, etc.), we provide a numerical illustration. The value of the parameters, which are mostly derived from Van Soest and Lensink [25], are as follows:

$$\begin{aligned} \bar{P} &= 45,000, \bar{P}_A = 150, \bar{Z} = 60, F_0 = 2000, \theta = 20, \\ \gamma &= 0.15, \alpha = 0.1, \beta = 0.1, \rho = 0.1, r = 0.2, \\ \Phi &= 17,000, T = 5. \end{aligned}$$

The steady-state values for deforestation, forest stock and shadow price of the forest are:

$$D^* = 343.42, \quad F^* = 1717.10, \quad \text{and} \quad \lambda^* = 25,204.$$

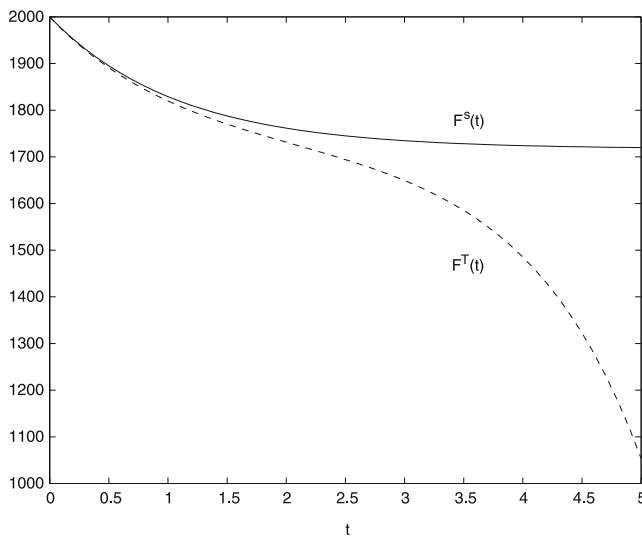


Figure 1. Forest stock time paths.

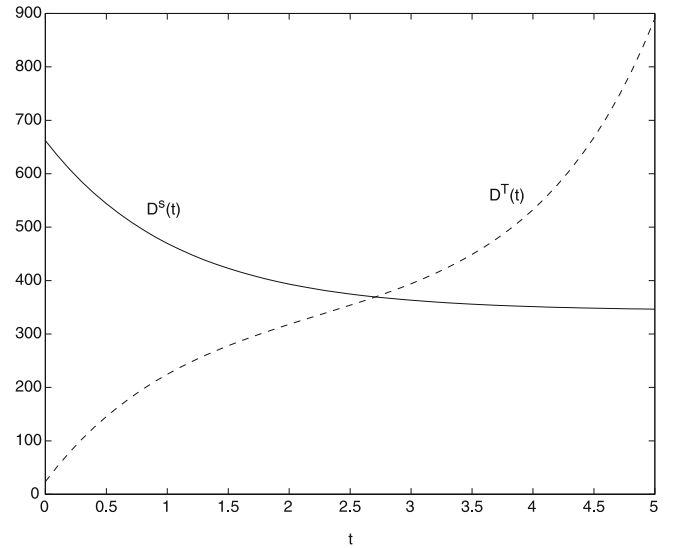


Figure 2. Deforestation time paths.

The time paths of the forest stock, the deforestation rate and the shadow price in the different scenarios are given by:

$$\begin{aligned} F^s(t) &= F^I(t) = F^* + 282.89e^{-0.93t}, \\ F^T(t) &= F^* + 286.84e^{-0.93t} - 3.94e^{1.03t}, \\ D^s(t) &= D^I(t) = D^* + 318.53e^{-0.93t}, \\ D^T(t) &= D^* - 322.97e^{-0.93t} + 3.26e^{1.03t}, \\ \lambda^s(t) &= \lambda^* - 18682e^{-0.93t}, \\ \lambda^T(t) &= \lambda^* - 18942e^{-0.93t} - 47.45e^{1.03t}, \\ \lambda^I(t) &= [24.431(1 - e^{-0.1(5-t)}) + 18.7e^{-0.1t} \\ &\quad (0.02 - e^{-0.83t})] \times 10^3 \end{aligned}$$

Figures 1–3 show these time paths. In these figures, continuous line corresponds to the sustainable scenario, while the dash line denotes the finite horizon scenario. These figures allow the following comments:

- In the sustainable scenario, the forest stock and the deforestation rate time paths decrease overtime towards their steady-state values (see figures 1 and 2). In the finite horizon scenario, the forestry country accelerates deforestation as the end of horizon approaches. Consequently, forest stock decreases sharply overtime. This result is expected in a finite-horizon optimization problem and justifies the idea of implementing an incentive policy to slow down deforestation.
- Figure 3 indicates that the shadow price of the forest stock is always increasing in both the sustainable and incentive scenarios. This is intuitive; as the forest stock decreases over time, the value of an additional unit of forest increases. Although the forest stock is also decreasing in the finite-horizon scenario, its shadow price is not monotonous. It is increasing in the beginning and decreasing on a final time interval. This last portion can be, again, explained by an end of horizon argument.

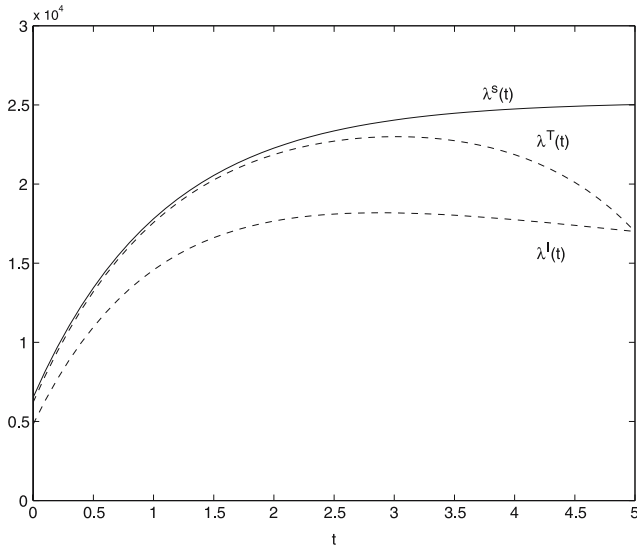


Figure 3. Shadow price time paths.

The revenue paths are as follows:

$$\begin{aligned} R^S(t) &= [-5.15e^{-1.85t} + 7.32e^{-0.93t} + 22.62] \times 10^6, \\ R^T(t) &= -29.93e^{0.1t} - 207.42e^{2.05t} + 91997e^{1.03t} + \\ &\quad [-5.30e^{-1.85t} + 7.42e^{-0.93t} + 22.62] \times 10^6, \\ R^I(t) &= R^S(t) + S(D^S(t), F^S(t)). \end{aligned}$$

It is easy to see that $R^T(t) > R^S(t)$, $\forall t \in [0, T]$. The total loss for the *South* if it applies the sustainable deforestation policy over the time interval is given by $J^T - J^S = 1.40 \times 10^6$. This total amount has to be compensated for by the *North* by a stream of transfers given by

$$S(D, F) = v(t)F(t) - w(t)D(t).$$

The last proposition showed that the above function can be constructed by first fixing $v(t)$ and next determining

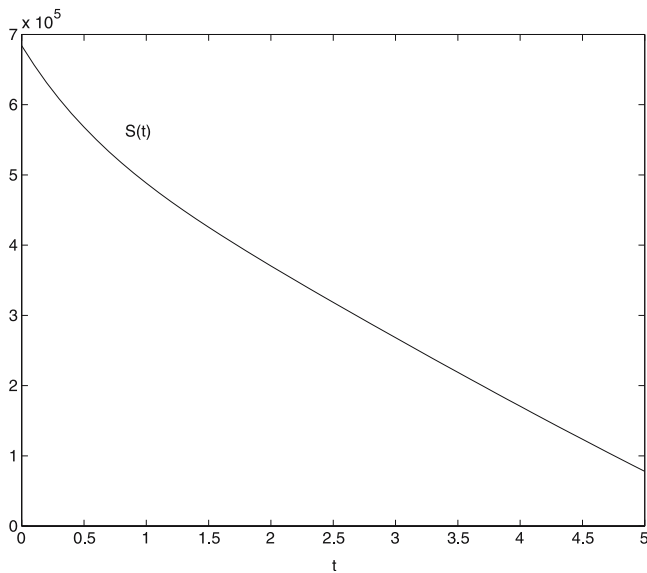


Figure 4. Transfers time path.

$w(t)$. To illustrate, let us consider a constant specification for function $v(t) = \bar{v} = 500$. It is easy to check that $w(t)$ would be given by

$$w(t) = -4522.8e^{-0.1t} + 5000,$$

and that it is positive for all t . The time path for the transfer function reads

$$S(t) = [-8.58 - 14.5e^{-0.93t} + 15.5e^{-0.1t} + 14.4e^{-1.03t}] \times 10^5.$$

The evolution of the transfers overtime is given by

$$\dot{S}(t) = [13.49e^{-0.93t} - 1.55e^{-0.1t} - 14.83e^{-1.03t}] \times 10^5,$$

which is negative. Thus the incentive policy in this example is to start supporting the *South* at a high level and decreasing it over time (see figure 4).

6. Conclusion

Using a detailed model showing the trade-off between the agriculture and forestry activities (deforestation and agriculture productivity), we showed that the international community can encourage forestry countries to participate in a program aiming at a better forest conservation while compensating it for the revenue loss it can bear from doing so. We also showed that by using incentive transfer mechanisms, the donor community or *North* can enforce a sustainable forest exploitation in the short run.

For further research, we can consider efforts to regenerate the forest. This would require a model incorporating a lag structure to handle correctly the link between replanting trees and later production of the resource. The *North* could also contribute financially to such a program in a principal-agent setting.

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Appendix

- Constants A_i , $i = 1, 2, 3$ in Proposition 1.

$$\begin{aligned} A_1 &= -\frac{(F_0 - F^*)^2(4a_1 + (2a_5 + a_3(2r + \Delta - \rho))(2r + \Delta - \rho))}{4}, \\ A_2 &= \frac{(F_0 - F^*)(F^*(a_5^2 - 4a_1a_3) + \lambda^*(a_5 + a_3(2r + \Delta - \rho)) + 2a_2a_3 - a_4a_5)}{2a_3}, \\ A_3 &= \frac{(a_4 - a_5F^* - \lambda^*)(a_4 - a_5F^* + \lambda^*) + 4a_3(-a_1(F^*)^2 + a_2F^* + a_6)}{4a_3}. \end{aligned}$$

- Constants C_i , $i = 1, 2$ and B_j , $j = 1, \dots, 6$ in Proposition 2.

$$C_1 = \frac{(\phi - \lambda^*)(2r - \rho + \Delta)a_3 + a_5 - (a_5^2 - 4a_1a_3)(F^* - F_0)e^{(\Delta + \rho)\frac{T}{2}}}{(a_3(\Delta + 2r - \rho) + a_5)e^{(\rho - \Delta)\frac{T}{2}} + (a_3(\Delta - 2r + \rho) - a_5)e^{(\Delta + \rho)\frac{T}{2}}},$$

$$C_2 = \frac{(a_3(\Delta - 2r + \rho) - a_5)(\phi - \lambda^*) + (a_5^2 - 4a_1a_3)(F^* - F_0)e^{(\rho - \Delta)\frac{T}{2}}}{(a_3(\Delta + 2r - \rho) + a_5)e^{(\rho - \Delta)\frac{T}{2}} + (a_3(\Delta - 2r + \rho) - a_5)e^{(\Delta + \rho)\frac{T}{2}}},$$

$$B_1 = -\frac{C_2(a_3a_5^2\Delta^2 + C_1(a_5^2 - 4a_1a_3)(-4a_1 + 2a_5(-2r + \rho) + a_3(\Delta^2 - (2r - \rho)^2)))}{4(a_5^2 - 4a_1a_3)^2},$$

$$B_2 = \frac{C_2^2(a_3a_5^2\Delta^2 + (a_5^2 - 4a_1a_3)(2(a_5 + 2a_1) + a_3(2r + \Delta - \rho))(2r + \Delta - \rho))}{4(a_5^2 - 4a_1a_3)^2},$$

$$B_3 = -\frac{C_2((a_5^2 - 4a_1a_3)\lambda^* + (2a_2a_3 - a_4a_5 + (a_5^2 - 4a_1a_3)F^*)(a_5 + a_3(\Delta + 2r - \rho)))}{2a_3(a_5^2 - 4a_1a_3)},$$

$$B_4 = \frac{C_1^2(4a_1 + (2a_5 + a_3(2r - \Delta - \rho))(2r - \Delta - \rho))}{4(a_5^2 - 4a_1a_3)},$$

$$B_5 = -\frac{C_1((a_5^2 - 4a_1a_3)\lambda^* + (2a_2a_3 - a_4a_5 + (a_5^2 - 4a_1a_3)F^*)(a_5 + a_3(2r - \rho - \Delta)))}{2a_3(a_5^2 - 4a_1a_3)},$$

$$B_6 = A_3.$$

- Constants M_i , $i = 1, 2$ in Proposition 3.

$$M_1 = \frac{C_1^2(4a_1 + (2a_5 + a_3(2r - \Delta - \rho))(2r - \Delta - \rho))}{4(a_5^2 - 4a_1a_3)} + \frac{(F_0 - F^*)^2(4a_1 + (2a_5 + a_3(2r + \Delta - \rho))(2r + \Delta - \rho))}{4},$$

$$M_2 = -\frac{C_1((a_5^2 - 4a_1a_3)\lambda^* + (2a_2a_3 - a_4a_5 - (4a_1a_3 - a_5^2)F^*)(a_5 + a_3(2r - \rho - \Delta)))}{2a_3(a_5^2 - 4a_1a_3)} - \frac{(F_0 - F^*)(F^*(a_5^2 - 4a_1a_3) + \lambda^*(a_5 + a_3(2r + \Delta - \rho)) + 2a_2a_3 - a_4a_5)}{2a_3}.$$

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